

Data structure and algorithm

Group Assignment

Target:-software engineering Second year second semester students

Section A and B

➤ **Write complete c++ programs to demonstrate the following question**

1. Create the following node(7,3,8,1 and 33) in circular doubly linked list
2. Create the following node(7,3,8,1 and 33) in circular single linked list.
3. Adding node to the end in circular doubly linked list.
4. Adding node to the end in circular single linked list.
5. Adding node to the front in circular doubly linked list.
6. Adding node to the front in circular single linked list.
7. Adding node to the left of a specific value in circular doubly linked list
8. Adding node to the left of a specific value in circular single linked list
9. Deleting node from the front in circular doubly linked list
10. Deleting node of specific data in circular single linked list.
11. Display a nodes in doubly circular linked list in forward and backward manner.
12. Consider the post-fix expression:
 $6\ 5\ 2\ 3 + 8 * + 3 + *$ design algorithm based previous postfix expression.
13. Adding node to the left of a specific value in circular single linked list
14. Adding node to the right of a specific value in circular double linked list
15. Display the list of value in circular single linked list or visit all node insert before.
16. Display the list of value in circular double linked list or visit all node insert before.
17. Write algorithm steps to Convert Postfix to infix expression.
18. Write algorithm to display the content of all elements in the stack in a backward manner in a linked list.
19. Write algorithm to implement the queue using singly linked list, assume we have two pointers head and tail, which point to the front and rear of the queue respectively.
20. write one sorting algorithm from(selection, insertion,bubble, merge) in descending order for list created before and which sorting algorithm works best.
21. Write c++ code that show the concept of Double Ended Queue in linked list.

Gp 1	Gp 2	Gp 3	Gp 4	Gp 5	Gp 6	Gp 7	Gp 8	Gp 9	Add Gp1	Add Gp1
1	2	1	1	2	2	1	2	2	1	2
3	4	4	7	4	6	5	4	7	12	8
5	6	5	9	8	8	7	6	11	14	12
7	8	7	14	13	13	11	8	17	16	15
9	10	11	17	19	15	16	10	20	17	17
16	12	14	18	20	20	20	12	21	21	21

Doing in group and submit before class end